Advisor meeting notes

Updated: 2026-10-27. Meeting date: October 26, 2026.

1. Checked how far representative price times volume is from observed revenue

The original representative price is the unweighted average of eligible UPC prices per mL. Compared this price times firm liquid volume with observed real revenue for the same eligible products.

Across 540 incumbent firm-months, January 2013-December 2023:

Measure	Result
Observed real revenue	$4.757 billion
Original representative price times volume	$7.162 billion
Aggregate overstatement	$2.405 billion (50.56%)
Median firm-month signed discrepancy	+35.17%
Full range of signed discrepancies	-42.98% to +186.06%

The original measure overstates revenue in 419 firm-months and understates it in 121. VUSE and JUUL contribute the largest dollar discrepancies.

2. Constructed a volume-weighted representative price that recovers revenue

I constructed a separate representative price using e-liquid volume as the UPC weights:

`p_volume_weighted = sum_u(q_u * p_u) / sum_u(q_u) = firm real revenue / firm liquid volume`.

Multiplying this price by firm liquid volume recovers observed eligible revenue.

3. Expanded demand instrument and price comparisons

The previously shown 48 specifications were four control sets x two firm-FE choices x two original brand-count instruments x three market-size definitions. The new exercise also varies price definition, instrument composition, sample availability, and whether quality is endogenous.

All results, rather than a selected subset, are included in the meeting results appendix. Each table states the price, controls, firm FE and sample, lists the complete excluded instrument set, and reports alpha and sigma (clustered SE) for all three market sizes. Endogenous-quality tables also report the quality slope. The specification index gives endogenous variables, excluded instruments, instrument count and overidentifying degrees of freedom for every run.

What varies

- **Price:** original UPC-unweighted price versus e-liquid-volume-weighted price.
- **Controls:** nicotine only; plus device size; plus log UPC variety; or both product controls. Nicotine remains in demand in every specification, with its exogeneity assumption varied below.
- **Firm FE:** with or without
- **Market size:** annual, monthly or constant purchasing intensity, using the existing definitions from last meeting.
- **Counts:** original national/geographic active-brand count (including fringe), annual incumbent count (>7% annual volume share), or monthly >7% count. New counts can replace an original count or supplement it; combinations are explicitly listed.
- **Lag quality:** previous calendar month's delivered nicotine, including pre-incumbency history when observed. Its role differs depending on whether current quality is treated as exogenous or endogenous.

Monthly sales by firms on the annual incumbent roster yield exactly the annual incumbent count here, because each roster member sells in all 12 months. The distinct monthly instrument instead reapplies the >7% threshold to each month's shares, without changing the demand roster.

Two different quality assumptions

Quality treatment	Jointly endogenous demand regressor	Example complete excluded instrument set	Identification
Exogenous	Price, within-nest log share	Tax + annual incumbent count	2 instruments / 2 endogenous: exact
Exogenous	Price, within-nest log share	Tax + active-brand count + annual incumbent count	3 / 2: one overidentifying restriction
Exogenous	Price, within-nest log share	Tax + annual incumbent count + lagged quality	3 / 2: one overidentifying restriction
Endogenous	Price, within-nest log share, current quality	Tax + annual incumbent count + lagged quality	3 / 3: exact

Quality treatment	Jointly endogenous demand regressors	Example complete excluded instrument set	Identification
Endogenous	Price, within-nest log share, current quality	Tax + active-brand count + annual incumbent count + lagged quality	4 / 3: one overidentifying restriction

Tax in every row is geographic population-weighted tax. All excluded instruments enter jointly; lagged quality is intended to help identify current quality in the endogenous-quality specifications, rather than being assigned exclusively to it. Current quality is an included exogenous control only in the exogenous-quality specifications. Tax plus lagged quality alone has too few instruments for three endogenous regressors and was not estimated with endogenous quality.

Scope and samples

Quality assumption	Sample	Instrument sets	Other combinations	Fits
Exogenous	528 complete-size rows / 132 months	9 without lag	4 controls x 2 FE x 2 prices x 3 market sizes	432
Exogenous	524 common-lag rows / 131 months	19, including nonlag baselines	Same	912

Quality assumption	Sample	Instrument sets	Other combinations	Fits
Endogenous	524 common-lag rows / 131 months	9 with lag and sufficient instrument count	Same	432
Total				**1,776**

These totals include deliberate sample-matched comparisons. The earlier 224 exploratory fits are not added to this total because they overlap and used a different sample rule for nicotine-only controls. All four control groups here retain the original complete-size sample rule. Common-lag models additionally exclude four January 2013 observations with no December 2012 quality. Existing BLU 2018 exclusions remain. Counts and market totals are constructed before these exclusions.

Interpretation and checks

357 of 1,776 runs have 0 < sigma < 1. There are enough that families are more informative than a list in the main notes. These counts include deliberate sample-matched repetitions; they are not independent successes, because they sometimes reuse the same data. The complete qualifying-run list records each specification, instrument list, identification status, sigma, SE and alpha. The family breakdown includes denominators.

The clearest pattern is the combination of volume-weighted price and an incumbent-count instrument, often without firm FE. The qualifying runs are not confined to one control set, one market-size definition, lag-quality models, or endogenous-quality models.

Price and firm FE family	Runs with 0 < sigma < 1 / all runs in family
Volume-weighted price, no firm FE	232 / 444
Volume-weighted price, firm FE	83 / 444
Unweighted price, no firm FE	35 / 444
Unweighted price, firm FE	7 / 444

- **Instruments:** 297 qualifying runs (successful sigma) include annual incumbent count (possibly with monthly count, active-brand count or lag quality); 55 include monthly >7% count without annual count; the remaining 5 use tax + lagged quality alone. No tax + original active-brand-count set qualifies, even when supplemented only with lagged quality, under either quality treatment. Every qualifying run therefore uses an incumbent-count instrument or the weak tax-plus-lag-only set.
- **Lagged quality is not required:** 171 qualifying runs omit it and 186 include it. The five qualifying tax-plus-lag-only models belong to the previously flagged weak-instrument family and should not be treated as a resolution on the basis of their point estimates.
- **Quality treatment:** 260 / 1,344 exogenous-quality runs qualify; 97 / 432 endogenous-quality runs qualify. All endogenous-quality runs include lag quality by construction, so this is not evidence that lag quality itself causes admissibility.
- **Controls:** qualifying counts are 64 / 444 for nicotine only, 80 / 444 with size, 90 / 444 with log variety, and 123 / 444 with both. Neither size nor variety is universally required.
- **Market size:** 118 / 592 annual, 133 / 592 monthly, and 106 / 592 constant-intensity runs qualify. No market-size definition is universally required.

Of the 357 qualifying sigma estimates, **354 also have a negative price coefficient; 3 do not**. Admissibility here refers only to the sigma point estimate: it does not require its confidence interval to lie within (0,1), a positive quality coefficient, or strong instruments. The full results should be checked for those features before selecting a specification.

All original 48 baseline coefficients and covariance matrices reproduce. New fits pass independent QR coefficient and structural-residual checks. Conditional F calculations account for two or three endogenous regressors; all three first stages are exported for endogenous-quality models. Original inputs and fits remain unchanged.

Overidentification df count restrictions; no overidentification test has been run. Neither an admissible sigma nor additional instruments establish exclusion. Lagged quality may correlate with persistent demand shocks; share-threshold counts depend on observed sales. Conventional intervals are not weak-IV-robust. No model selected and no sigma clipping.

4. Reran nested logit at the firm–state–month level

Question: Does moving from national firm-month aggregation to firm–state–month observations change the demand estimates?

Completed **504 specifications** on a common sample of **24,674 incumbent firm–state–month observations, 51 jurisdictions (50 states plus DC), and 131 months**. The grid varies four control sets, two price definitions, three FE structures (pooled, additive brand/state FE, or brand-state pair FE), three existing consumption-intensity variants, and instrument sets with exogenous or endogenous current quality. Exogenous-quality models use five sets: tax plus local active-brand count, national annual incumbent count, local monthly >7% count, or either incumbent count plus lagged quality. Endogenous-quality models use the two count-plus-lag sets with enough instruments.

What changes at the state level

Component	State-level construction
Tax instrument	State's real tax per mL, replacing the geographic population-weighted tax
Representative price	Equal-UPC mean of local UPC prices, or local real revenue / local liquid volume
Quality	Equal-UPC delivered nicotine among products sold locally
Device size	Strict equal-UPC mean device size among locally sold device UPCs; a given UPC's physical capacity stays unchanged
Variety	Log number of locally sold eligible UPCs
Shares	Firm's local liquid volume relative to state market size; within-nest share uses all eligible local brands, including fringe
Market size	Existing national market size multiplied by the state's adult population share; national NCP consumption intensity retained
Counts and lag	Local active-brand and monthly >7% counts; national annual >7% incumbent count retained; previous-calendar-month quality constructed within brand-state

The national annual incumbent roster remains the demand roster, with positive local sales required. UPC eligibility retains the national three-store cutoff; no additional state-specific store cutoff is imposed. All specifications use the same complete-size and complete-lag sample. Standard errors are clustered by state and month. No state-month FE are included because of the single-nest identity.

Findings

For **both size and variety controls, additive brand and state FE, annual-intensity market size, exogenous quality, and state tax + local active-brand count**:

Measure	Equal-UPC price	Volume-weighted price
Price coefficient alpha (SE)	−1.176 (1.415)	−0.707 (0.375)
Sigma (SE)	2.274 (2.325)	0.485 (0.274)
Tax coefficient in price first stage (SE)	+0.345 (0.221)	+0.397 (0.180)

The volume-weighted sigma point estimate is inside (0,1), but its approximate 95% interval is **−0.07 to 1.04**, extending beyond both boundaries. The very large SE of 2.325 belongs to the equal-UPC sigma estimate. The tax–price first-stage slopes are positive at this level, although the equal-UPC slope is imprecisely estimated.

Volume-weighted sigma is not usually admissible across the full state-level grid: it lies strictly between 0 and 1 in **91 of 252 runs (36%)**; 89 also have negative alpha. For equal-UPC prices, the corresponding counts are **82 of 252 (33%)**, with 80 also having negative alpha. These are sensitivity counts across related specifications, not independent evidence or a model-selection rule. Some estimates are highly unstable; admissibility depends on instruments, controls, FE, and price construction.

Interpretation: moving to state markets changes the estimates substantially, and some volume-weighted specifications have admissible sigma point estimates. This does not establish national aggregation as the cause of sigma >1. Local assortment, shares, market-size allocation, and count instruments change together; a matched national/state comparison is needed to isolate their contributions. State market sizes assume common national consumption intensity, and neither instrument validity nor adequate identification is established. Conventional SEs and intervals are not weak-IV-robust.

State UPC totals reconcile to canonical national volume, units, and revenue within 1e-8 relative tolerance. Independent residualized QR coefficient checks passed for all 42 size-plus-variety/annual-market specifications (maximum discrepancy below 3e-8); source input hashes are unchanged. Existing national demand estimates and canonical inputs were preserved.

Results: findings, all specifications, coefficients and clustered SEs, and construction details. Scripts: `code/3_demand_estimation/` and `22_validate_firm_state_month.R`.

5. Advisor discussion and agreed next steps

Recorded October 27 following the October 26 meeting. These decisions guide the next stage; the preceding sections record the investigations presented at the meeting.

1. **Proceed with sales-weighted prices for now.** Use the liquid-volume-weighted representative price discussed above.
2. **Proceed without firm fixed effects for now.** One hypothesis discussed was that the available observations and within-firm variation may be insufficient once firm FE are included. Firm FE may absorb too much relevant variation, contributing to unstable results in which inside share appears to predict outside share. This is a possible explanation to investigate, not an established cause.
3. **Choose a provisional national firm-month specification to continue to the supply side.** For now, deliberately select a specification with a reasonable sigma, even if its conditional price–tax relationship is negative. Revisit the demand specification and the negative price–tax relationship later. This is a pragmatic interim choice; a particular specification has not yet been selected in these notes.
4. **Explore additional instruments later.** With firm FE omitted, consider Gandhi–Houde instruments and assess whether they also help resolve the sigma issue.
5. **Tax aggregation remains flexible.** As discussed, taxes may use volume weights or the original unweighted UPC simple average. This records the advisor's comment; the precise UPC-level tax construction remains to be specified, since the existing national tax instrument uses geographic population weights rather than an unweighted UPC average. No alternative tax series was constructed as part of this discussion record.